KnowSelf

Agentic Knowledgeable Self-Awareness for LLM-Based Agents

Teaching LLM Agents When to Think, Reflect, or Seek Knowledge

arxiv.org/abs/2504.03553

ML Researchers & Graduate Students | Interactive Planning & LLM Agents

Current Agent Training Uses 'Flood Irrigation'

Indiscriminate Knowledge Injection Hurts Efficiency

Trajectory Imitation

ReAct

Blindly copies gold trajectories without situational assessment

Trial-and-Error

Reflexion

Refines through feedback without situational awareness

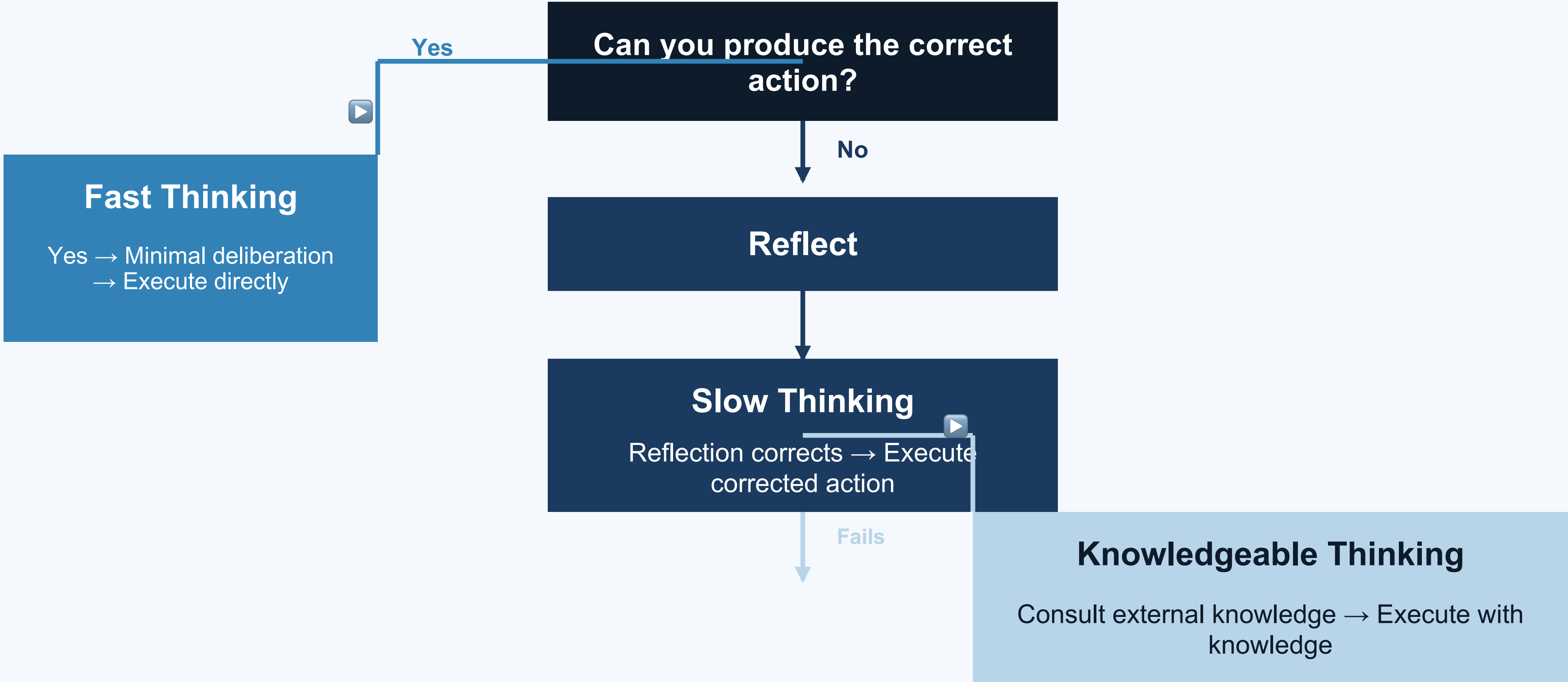
Knowledge-Augmented

KnowAgent, WKM

100% knowledge injection regardless of actual need

Humans possess metacognitive self-awareness — agents need the same

Three Situational Thinking Modes



KnowSelf Framework: Data → Training → Inference

1

Data Construction

- Self-explore environments
- Heuristic classification
- Special tokens mark situations

2

Two-Stage Training

- SFT (Supervised Fine-Tuning)
- RPO (Reward-based Policy Optimization)

3

Self-Aware Inference

Special tokens trigger:

- Direct action
- Reflection
- Knowledge retrieval

Lightweight heuristic labeling — no expensive human annotation

KnowSelf Correctly Handles Tasks Where Baselines Fail

Task: "Put a hot mug in cabir

OpenAI-O1

X Incorrectly takes egg from microwave

KnowSelf (Llama-8B)

✓ Uses Reflection token to realize it should heat the mug using the open microwave

Reflection

Task: "Put a saltshaker in dra

DeepSeek-R1

X Tries to close drawer first (which fails)

KnowSelf

✓ Applies Knowledge token noting that agent should place object without removing unrelated items

Knowledge

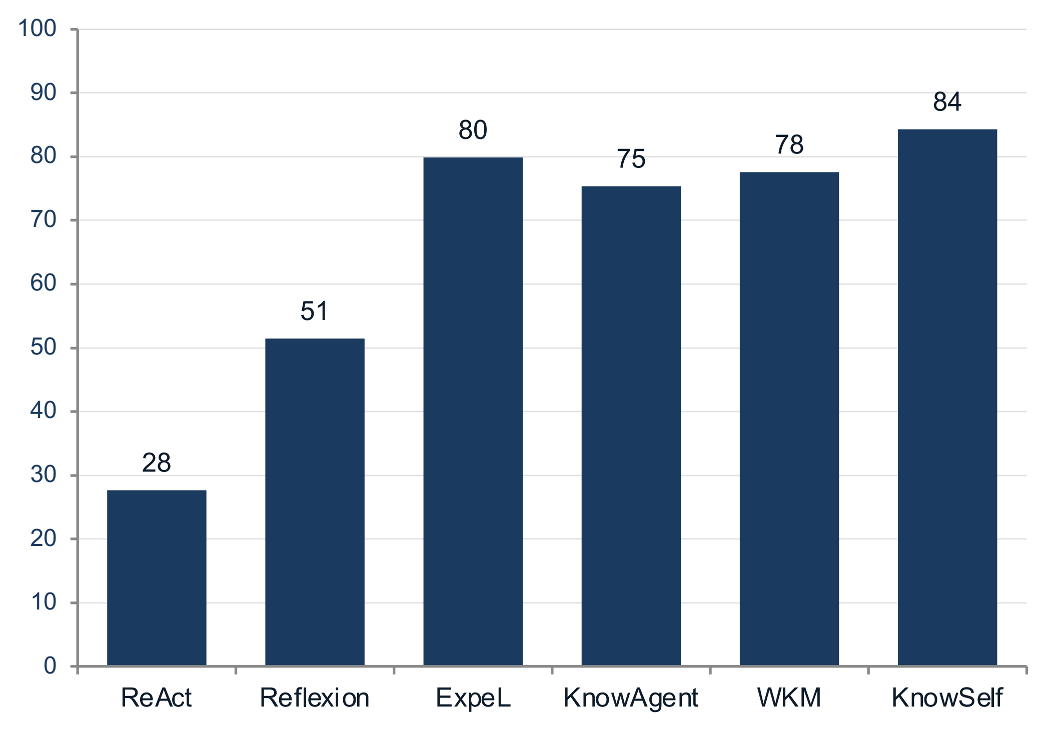
KnowSelf Achieves 84.33% on ALFWorld with Only 15% Knowledge Usage

Backbone	Method	Know%	Put	Clean	Heat	Cool	Examine	Put Two	All
GPT-4o	ReAct	0%	83.33	74.19	69.57	85.71	77.78	64.71	76.12
GPT-4o	Reflexion	0%	100.00	87.10	73.91	90.48	83.33	70.59	85.07
GPT-4o	ExpeL	**100%**	95.83	83.87	69.57	80.95	88.89	52.94	79.85
Llama-8B	ReAct	0%	33.33	3.23	0.00	57.14	66.67	23.53	27.61
Llama-8B	Reflexion	0%	62.96	22.73	5.88	64.29	86.36	50.00	51.49
Llama-8B	ExpeL	**100%**	83.33	32.26	30.43	23.81	55.56	17.65	41.04
Llama-8B	ETO	0%	91.67	70.59	82.61	61.90	88.89	64.71	78.36
Llama-8B	KnowAgent	**100%**	87.50	93.55	65.22	66.67	61.11	64.71	75.37
Llama-8B	WKM	**100%**	95.83	87.10	86.96	61.90	66.67	52.94	77.61
Llama-8B	**KnowSelf**	**15%**	**91.67**	**87.10**	**91.30**	**85.71**	**77.78**	**64.71**	**84.33**
Gemma-2B	ReAct	0%	0.00	9.68	0.00	4.76	44.44	0.00	8.96
Gemma-2B	ETO	0%	91.67	83.87	78.26	52.38	77.78	29.41	71.64
Gemma-2B	KnowAgent	**100%**	91.67	90.32	69.57	71.43	66.67	41.18	73.88
Gemma-2B	WKM	**100%**	91.67	87.10	78.26	71.43	61.11	52.94	76.12
Gemma-2B	**KnowSelf**	**26%**	**87.50**	**93.55**	**73.91**	**76.19**	**83.33**	**52.94**	**79.85**

84.33% > GPT-4o ExpeL 79.85% with 15% vs 100% knowledge

ALFWorld benchmark results

Minimal Knowledge, Maximum Impact



KnowSelf achieves best accuracy with 6x less knowledge

KnowSelf (Llama-8B)
15% knowledge **84.33% acc**
Best accuracy

ExpeL (GPT-4o)
100% knowledge **79.85% acc**
6x more knowledge

KnowAgent / WKM
100% knowledge **75-78% acc**
6x more knowledge

Key Takeaways

1. Indiscriminate knowledge injection is wasteful — selective use outperforms blanket application
2. Agents can learn situational self-awareness via special tokens with lightweight heuristic labeling
3. KnowSelf Llama-8B (84.33%) outperforms GPT-4o ReAct (76.12%) and GPT-4o ExpeL (79.85%)
4. Only 15-26% knowledge usage needed vs 100% for competing methods
5. Gemma-2B reaches 79.85% — approach generalizes across model scales

Source: arxiv.org/abs/2504.03553